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A Protocol for Multibiometric Data Acquisition, Storage and Dissemination

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CONTENTS

I	INTRODUCTION	3
II	MULTIMODAL BIOMETRIC DATABASES	4
III	DATA COLLECTION	7
III-A	Acquisition systems and protocols	8
III-B	IRB	12
III-C	WVU Biometric Data	12
IV	BIOMETRIC DATA STORAGE	13
IV-A	BIOMDATA DATABASE DESIGN	15
IV-A.1	Description of data for individual modalities	19
IV-B	DATA COLLECTION PACKAGE FOR BULK UPLOAD	21
IV-B.1	COLLECTION database design	21
IV-B.2	Graphical User Interface for COLLECTION database	22
IV-B.3	Biometric Scanner Interface	22
IV-B.4	Generate XML files	22
IV-B.5	Bulk upload programs	24
IV-B.6	Naming conventions for the biometric image and sound files	24
V	DISTRIBUTION	26
VI	CONCLUSIONS and PERSPECTIVES	26
	References	26

I. INTRODUCTION

Biometrics is the science of identifying an individual based on the physical or behavioral characteristics such as fingerprints, iris, face, signature, voice, etc. It is a rapidly growing field that encompasses the areas of computer science, electrical engineering, forensics and statistics, with plenty of opportunities for research, that is fueled by the need for higher security and efficient usage. The various stages of a generic biometric system consists of four modules [10]:

- *Sensor module* captures the biometric data. An example is a camera used to capture the person's face.
- *Feature extraction module* processes the biometric information and builds the template as a set of distinct extracted features. An example is the location and shape of the facial attributes such as eyes, lips, nose, chin and their spacial relationships.
- *Matcher and decision making module* compares the template extracted during the recognition process with the own template stored in the database. As a result of the comparison a match score is generated, based on which the decision for the identity of the person is made.
- *Database system module* stores one or more templates generated during the enrollment process of an individual. Before performing the biometric recognition, the person has to be enrolled in the system. In the enrollment phase, biometric data is captured with the sensor, a quality check of the data is performed to ensure that features can be extracted in the next processing steps. Then a reference measure based on features extracted from enrollment samples is stored in the database.

The performance of a biometric system is evaluated through the error rates:

- FMR - false match rate, named also false accept rate when the biometric measurements from two different persons are mistakenly labeled as from the same person.
- FNMR - Non-false match rate known also as false reject rate, when the biometric measurements from the same person are mistakenly considered to be from two different persons.

These two error rates depend on the threshold t chosen for the biometric system. Threshold t is the boundary between the overlapped impostor and genuine score distributions. The genuine distribution is the distribution of scores generated from pairs of samples from the same person, and the impostor distribution is the distribution of scores generated from pairs of samples from different persons. If the resulting match score obtained from the presented biometric is higher than the threshold t then the person is identified as a genuine, otherwise is identified as an impostor. The point at which the FAR and FRR meet or crossover is called equal error rate EER.

- FTC - failure to capture, refers to the percentage of times the device fails to capture the biometric when presented.
- FTE - failure to enroll, refers to the percentage of times the device is not able to enroll the user in the biometric device.

These two error rates depend on the ability of the device to localize the biometric and on the quality of captured biometric

characteristic when presented.

Biometric systems can be classified as unimodal or multimodal. Examples of existing unimodal biometric databases are gathered in Table I. Unimodal systems rely on a single source of biometric information unlike multimodal systems that utilize more than one source of biometric information. It is demonstrated that multimodal biometric systems are more accurate and reliable compared with unimodal biometric systems. The fusion of multiple biometric sources provides more information in the process of identification and are more robust against impostor attacks, since multiple human traits are difficult to forge. The multiple sources of biometric information can be categorized as follows [18]:

- *Multiple sensors* that capture the same biometric feature. Example includes the use of optical, electro-optical, capacitive or ultrasound fingerprint scanners that can capture the same fingerprint.
- *Multiple units*, such as two irises, one from each eye, or two fingerprints from two different fingers.
- *Multiple algorithms*, such as using multiple feature extraction and/or matching techniques on the same fingerprint samples.
- *Multiple traits*, where two or more biological characteristics, such as face and fingerprint are used.

This raises the need of *organized multimodal biometric research databases* with established procedures and protocols in data collection, in order to avoid data collection errors. Multimodal biometric databases are needed to evaluate the feature extraction, template matching and decision making algorithms used by biometric devices and also in performance measure, testing and evaluation of biometric systems [11].

II. MULTIMODAL BIOMETRIC DATABASES

Examples of existing multimodal biometric databases are gathered in Table II.

Biometric researchers need to have access to a centralized, reliable and accurate, reusable and high secure, web-enabled multimodal biometric database, that can be easily incorporated in their study and work.

- Centralized and web-enabled refers to the availability of biometric data to all researchers' sites in real time, addressing issues as data transfer, replication and integrity.
- Reliable and accurate. Biometric data collected has to be associated with the information related to the environmental conditions under which it was collected, with biometric devices used when collected, with protocols adopted and with information about itself (resolution, file type etc.). Using the same multimodal biometric database makes it easier to evaluate different biometric technologies.
- Multimodal. Multiple biometric characteristic for one person are stored in the database that are connected through a PIN or random identification number. Such databases are also a data source for unimodal biometric research.

TABLE I

EXAMPLES OF UNIMODAL BIOMETRIC DATABASES

Database name	Biometric	# subjects	Samples/subject	Organization
FVC2004 [1]	Fingerprint (DB1, DB2 DB3)	90	2 fingers 4 impressions/finger 3 sessions	Univ.of Bologna Michigan State Univ. San Jose State Univ.
M2VTS [1]	Face	37	1 shot/person 5 sessions	Laboratoire de telecommunication et teledetection
	Voice	37	1 phrase of 10 words 5 sessions	
YOHO [1]	Voice	138	4 enrollment sessions/subject 24 phrases /enrollment session 10 verification sessions/subject 4 phrases/verification session	Linguistic Data Consortium
DB NIST [12][1]	Fingerprint (3 DBs, 1DB/fingerprint scanner)	90, 30/DB	4 fingers/subject 4 impressions/finger, 3 sessions variable pose	National Institute of Standards Technology
	Face (MID) (FERET)	1573 200 to 1010	2 images/subject to 26 images /subject	
ELRA [1]	Voice multiple DBs SpeechDat Babel etc.	adults, children over telephone radio,internet television	variety of languages from specialized domains	European Language Resources Association
Notre [14] Dame	Face	multiple collections 82 to 334	variable pose 159 to 17856 images	Univ.of Notre Dame
	Ear	114 to 302	multiple collections 464 to 2245 images	
	Hand	223	2382 images	

- Reusability refers to the raw data that undergoes processing operations, and then by sharing, time and computational resources are saved (deblurring of face images, preprocessing of fingerprint images for dirt, dryness, or moisture etc.).
- High secure refers to a variety of methods to secure the stored biometric data, through passwords, encryption, using different levels of access into the database through roles and privileges, transactions tracking and watermarking.

In comparison with other multimodal biometric databases, our multimodal biometric database incorporates a higher number of biometric modalities. Through a random identification number assigned to each subject, images of face, fingerprint, iris, hand, palmprint and sound of voice can be linked together.

The novelty of our paper consists in the fact that it gathers information not only about the multimodal biometric data collected, but also about the architecture and configuration of an entire system to store such information, using relational database management systems. Besides the primary biometric identifiers, we collect "soft" biometrics that augments the identity information and can be used to improve the search speed by filtering a large biometric database [8] [9]. The identification or verification performance relies strongly on the acquired data that further depends on the sensors used, the environment in

TABLE II

MULTIMODAL BIOMETRIC DATABASES

Database name	Biometric	# subjects	Samples/subject	Organization
MyIdea [3][4]	Fingerprint Video Face, Voice Signature, Voice Handwriting, Voice Palmprint Hand geometry	104	3 sessions 6 images/finger/session 10 fingers, 2 sensors different scenarios 6 signatures different scenarios different scenarios 4 images/right hand 3 images/right hand	Univ. of Fribourg, Switzerland, Engineering School of Fribourg, Switzerland, Groupe des Ecoles des Telecommunications Paris
MCYT [16][2]	Fingerprint Signature	330	10 fingers 12 images/finger 2 sensors 25 samples own signature 25 skilled forgeries	Universidad Politecnica de Madrid Univ. of Valladolid Univ. of the Basque Country Escola Politecnica de Mataro
BIOMET [5]	Face, Audio Face infrared Hand Fingerprint Signature Face 3D	327 91	3 campaigns/one shot 16 images/person 5 images/person 2 fingers 6 images/finger/sensor 15 genuine/person 17 impostor/person 5 shots	BioSecure

which data is acquired, the appearance and behavior of the subjects etc. This information is also maintained in our database and plays an important role in biometric research [17][13](explained in Section IV-A). Information in the database is organized on three levels: raw data, templates and match scores.

The system that we describe is referred to as the Joint Archived Multimodal Biometric Dataset Collection, Figure 1. This system includes the following modules:

- Biometric data collection: modalities, acquisition systems, description of protocols and procedures, described in Section III-A. This biometric dataset is meant to aid researchers in their work, to develop, train, test and evaluate the human recognition algorithms. The algorithms used in processing the data have a major influence in biometric recognition. This suggested us to broke down the data storage in different levels of data such as raw data, templates and match scores.
- Biometric data storage, that refers to the design and implementation of a relational management database system. The advantage of storing the biometric data in the database is based on the inherent advantage presented by database systems: centralized, accurate, highly secure, reliable and organized data, easy access to insert, update and retrieve biometric data,

described in Section IV.

- Access to the biometric dataset: distribution using CDs, DVDs, download via Internet or registration and other types of access granted to the database.

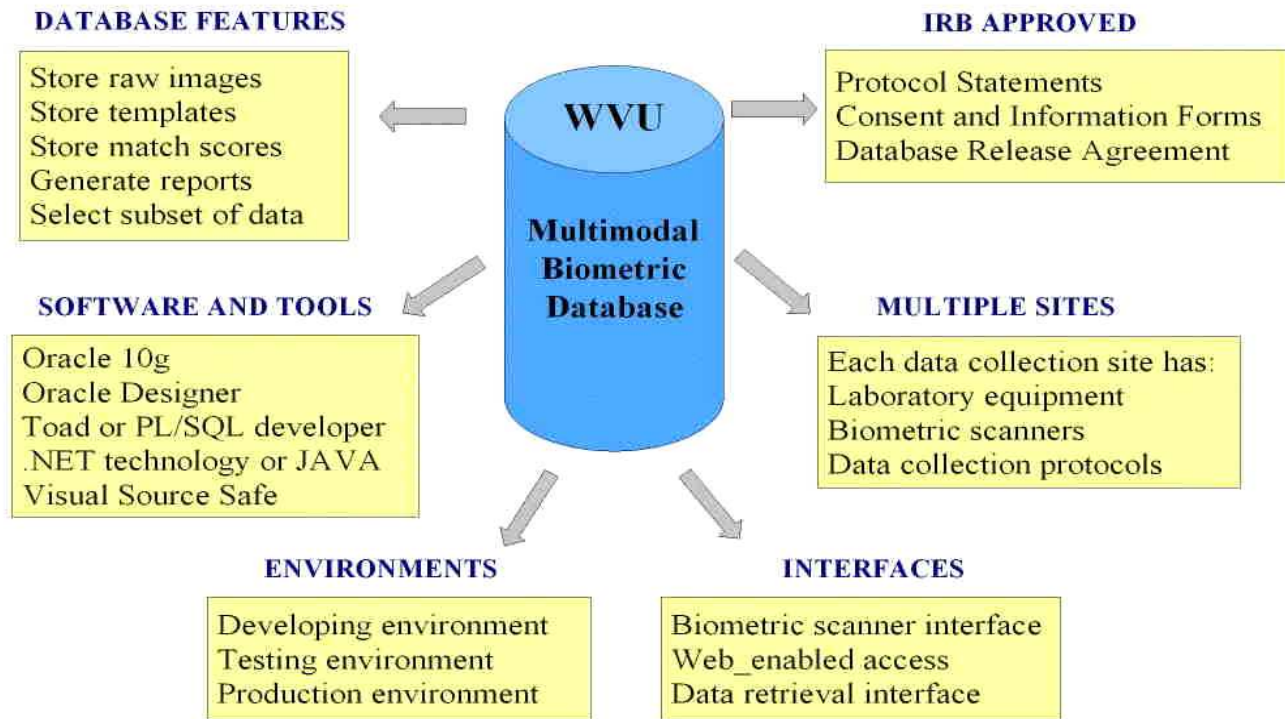


Fig. 1. Design of the Joint Archived Multimodal Biometric Database

The paper is divided as follows: Section III describes the biometric data collection: acquisition systems and protocols for every modality, IRB, samples of data for each modality, Section IV describes the biometric data storage, architecture and configuration of the Oracle 10g database and the bulk upload package that we developed.

III. DATA COLLECTION

The first available biometric recording sets collected at West Virginia University, Multimodal Biometric Dataset Collection Laboratory, Figure 3 a, b, includes face, fingerprint, palmprint, iris, hand geometry and voice data. Besides biometric data, we collect additional biographic information that is described in Section IV-A.

A. Acquisition systems and protocols

The data collection protocol is divided in three sections. An instance of biometric data collection for a subject is shown in Figure 2.

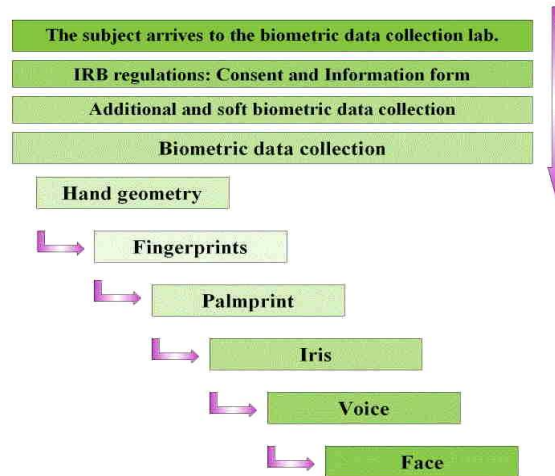


Fig. 2. An instance of biometric data collection at WVU laboratory

1) IRB regulations

The subject is explained the multimodal biometric data collection and encouraged to ask questions about it. The data collector reviews all subject's questions and concerns. Then the subject is asked to read carefully and sign the IRB consent and information form. The original copy is filed in a locked and secure cabinet file, and a copy is given to the subject.

2) Additional and Soft Biometric data collection

If the subject is at the first session, he/she is assigned a unique random identification number; if it is a repeat, meaning that he/she already had a first session, the data collector checks if it has been at least 2 weeks since the date of his/her last collection. The subject provides information related to age, gender, ethnicity, his/her appearance etc., information that is described in more detail in Section IV-A; measurements of subject's height and weight are taken using a regular digital weight scale and a SECA 214 Portable stadiometer; environment temperature and humidity are also registered, see Figure 4 g.

3) Biometric data collection

In every session, the collection begins in a random order. When multiple scanners are used for one modality in the same session, the order of the scanners is chosen randomly for each subject.

- For the *face* modality, frontal images are collected indoor in a controlled scenarios, with uncluttered, uniform

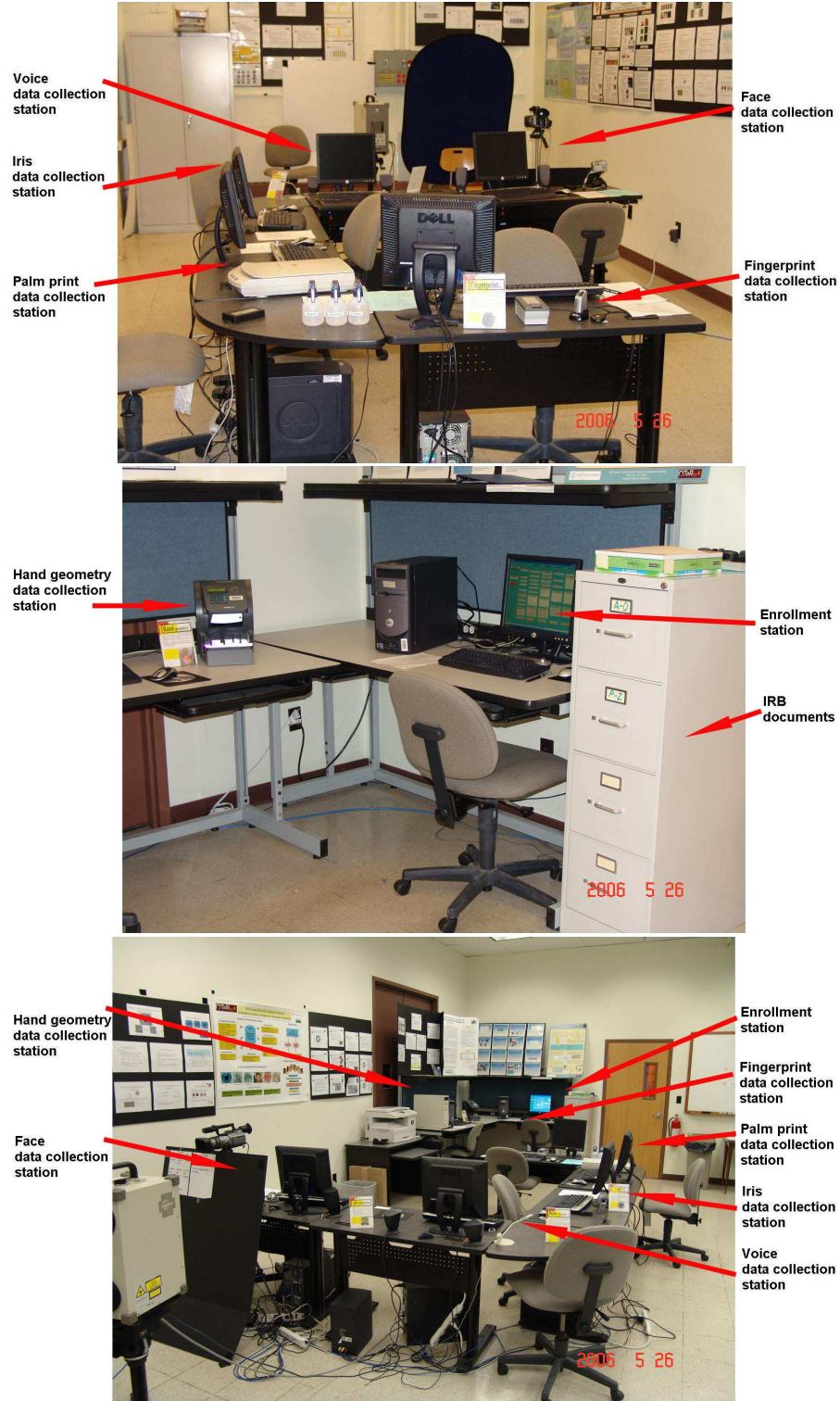


Fig. 3. WVU Multimodal Biometric Data Collection Laboratory; photos of the lab from different angles presenting biometric data collection stations

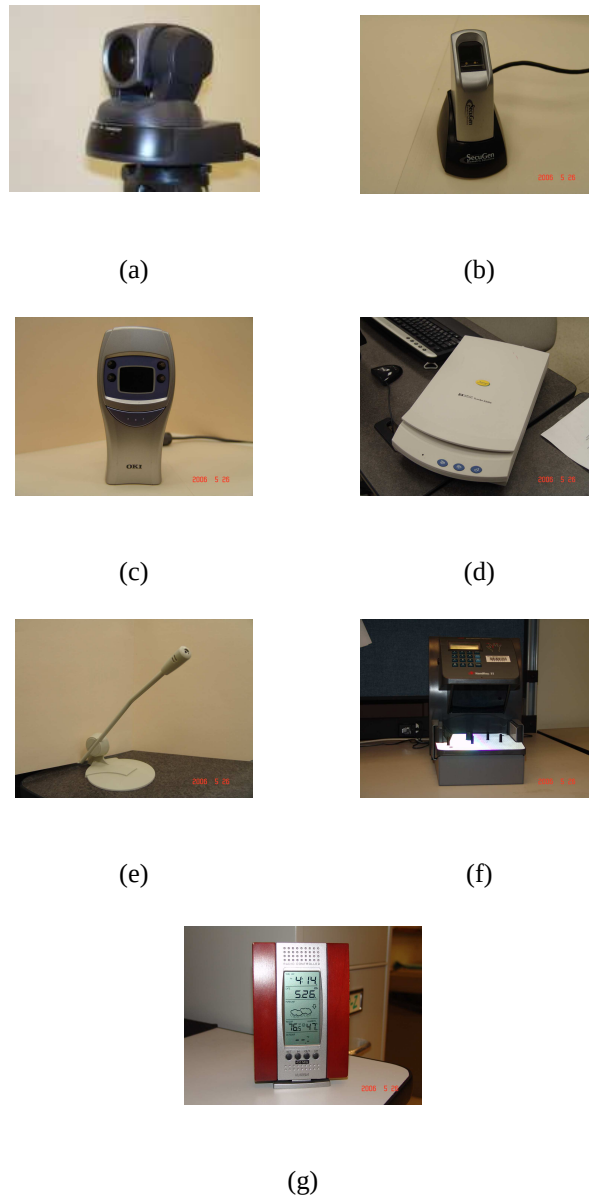


Fig. 4. Biometric devices: (a)Sony DV30/31 (b)SecuGen fingerprint scanner (c)OKI IRISPASS-h (d)HP Scan Jet 4200C (e)Microphone (f)IR Recognition Systems HandKey II (g)Device to measure temperature and humidity

background. The subject is positioned at a distance of 2 meters from the camera, and directed to look into the camera; no emotions are specified; eyeglasses are not removed. The only light used is that from the ceiling (i.e., ambient "office" lighting). A SONY EVI D30/31, using IC Standard Capture software captures the image, Figure 4 a. The size of the images is 1.26 MB, 576x768x3 or 900 kB, 480x640x3.

Using zoom in, the subject is framed from shoulders to top of the head or until face (from chin to hair) fits into the picture window. The subject is asked to stand up and sit back down again before the second picture is taken. This process is repeated until 5 images of face are obtained. See images from Section III-C, Figure 6.

- For the *fingerprint* modality, the hardware that we used to collect the fingerprint images is the optical fingerprint

biometric scanner SecuGen, Figure 4 b, 500 dpi [19] and the software is our own implementation in JAVA programming language that allows us to collect single or time series images. Fingerprints are collected without controlling the quality and the centering of the finger and without cleaning the glass plate of the scanner between acquisitions. The image size is 72 kB, 292x248.

All fingers have a code notation L for those from the left hand and R for those from the right hand. Fingers are numbered from 1 to 5 starting with the thumb and finishing with the pinkie, Figure 5. Images of the thumb and index fingers from both hands are taken in the following order: L1, R1, L2, and R2. The subject is instructed to lift and replace her finger between each capture. Five images of each finger are taken. See images from Section III-C, Figure 9. No chemical substance is used to clean the fingers, prior to data acquisition.

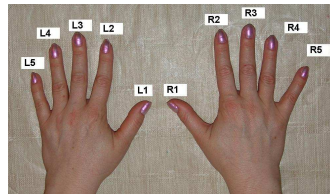


Fig. 5. The code notation for the fingers

- For the *iris* modality, the user is asked to hold the device away from one eye (at a distance eye glasses would be away from the face) while covering the other eye with the hand; the covered eye must remain open, so that the pictured eye does not squint. The user should be able to see all sides of the green box displayed on the screen of the OKI IRISPASS-h handheld device [6], Figure 4 c. The size of the picture is 302 kB, 480x640 and 2 kB for the log file. The log file contains information about the device settings when data is collected, such as focus value, specular height, aperture value etc. Four images of each eye are taken. At the discretion of the volunteer more data samples were acquired when subjectively was determined that the quality of the image was poor. See images from Section III-C, Figure 8.
- For the *voice* modality, data is captured twice, resulting in two files with extension .wav, using a standard sound recorder Logitech that does not filter out the background noise. The subject speaks at about 6 inches from the microphone, Figure 4 e and is asked to read a text composed of two short sentences "My favorite sport is basketball. I love to support this research", and one digit numbers "1, 6, 7, 2, 3, 0, 8, 7, 4, 5, 9, 1". The files are typically 10 to 25 seconds in length.
- For the *hand* modality, the subject is first enrolled into the system with the right hand using his/her random identification number as the pass code. The subject is instructed how to enter his/her random identification number as a pass code in the biometric device and place the hand on the platen. Then the subject collects by himself the

right hand image for six times. The right hand image is captured using an IR Recognition Systems HandKey II [7], Figure 4 f and a Matlab program is used to read the top view image of the hand from the RSI to the computer, see images from section III-C, Figure 7. The size of the image is 20 kB, 660x768 and 2 kB for the .txt file created at the moment of data collection that contains feature vectors and match scores.

- For the *palmprint* modality, the subject is instructed verbally how to place the hand inside the scanner without touching the glass or place lightly the hand on the screen, without applying any pressure on the glass. The fingers are spread naturally. Rings are not removed. One image is captured for each palm with the lid closed using the HP ScanJet 4200C, Figure 4 d. The image size is 5.95 MB, 1730x1276x3. See images from section III-C, Figure 10. The platen is cleaned from time to time with alcohol.

B. IRB

The purpose of West Virginia University - Institutional Review Board (IRB) for Protection of Human Research Subjects is to protect the rights and welfare of individuals who serve as subjects in the research conducted by faculty, staff and students and to ensure institutional compliance with accepted ethical standards.

The IRB applies the guiding principles and establishes policies when dealing with human subjects, reviews periodically all research projects to check if procedures do not violate the safety, health or life of those subjects. These guiding principles refer also to data collection and continue with data manipulation and analysis and establish the requirements and restrictions for biometric data release and publication.

Depending on the nature of the research project, documents submitted for approval, the I.R.B. encompasses procedures used, human subject details, risks and discomforts, financial considerations, benefits, confidentiality, advertisement, items of special concern, investigators and contact person information.

W.V.U. biometric dataset is collected based on I.R.B. approved documents that comply with the guidelines described by I.R.B. Every session, unconstrained consent is obtained from the subjects that participate in the study. Members of the research community involved in biometric research and interested in WVU biometric data will sign and comply with the specifications in the *Database Release Agreement*.

C. WVU Biometric Data

The first records of our multimodal biometric data collection available include face, iris, fingerprint, voice, palmprint and hand geometry from subjects of different age, gender and ethnicity with variable number of sessions and samples/session as described at the protocols. If the subject was reluctant to a biometric modality, we did not collect his/her biometric trait for

TABLE III

WVU BIOMETRIC DATA

Biometric modality	# of subjects	# of files
Iris	244	3099 bmp 3099 log
Fingerprint	272	7219 bmp
Palm	263	683 bmp
Hand	217	3062 jpg 3062 log
Voice	274	714 wav
Face	112 206	566 bmp 768x576 1181 bmp 640x480

that modality, therefore the number of subjects/modality is different.

Figures 6, 7, 8, 9 and 10 depicts biometric images collected in our laboratory.



Fig. 6. Face examples

IV. BIOMETRIC DATA STORAGE

The core of the multimodal biometric dataset collection project is an Oracle 10g database called "BIOMDATA" described in Section IV-A. The advantages of storing the biometric data into a relational database management systems were explained in Section II. As shown in Figure 11, we envision two ways of populating the database:

1) BULK UPLOAD PACKAGE

We developed a Data Collection Package, Section IV-B, that consists of:

- a) An Access database named "COLLECTION" database, Section IV-B.1

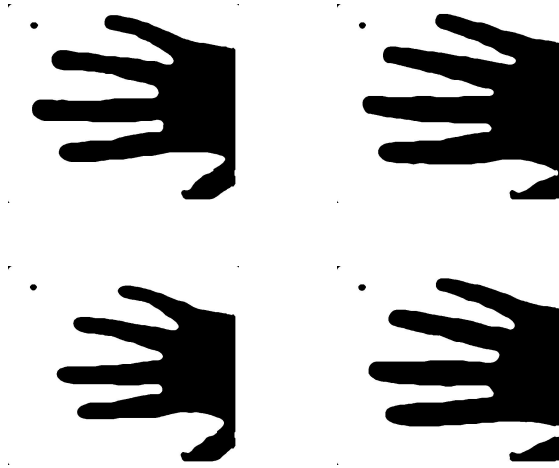


Fig. 7. Hand geometry examples

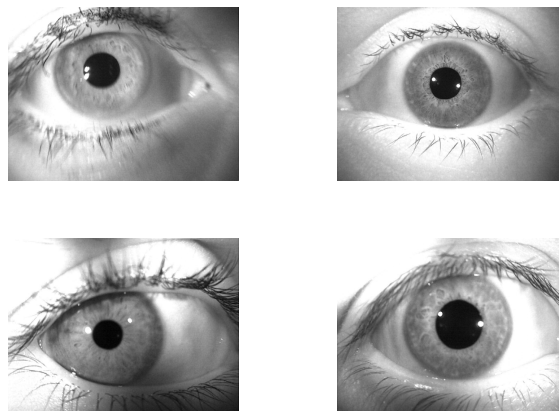


Fig. 8. Iris examples



Fig. 9. Fingerprint examples

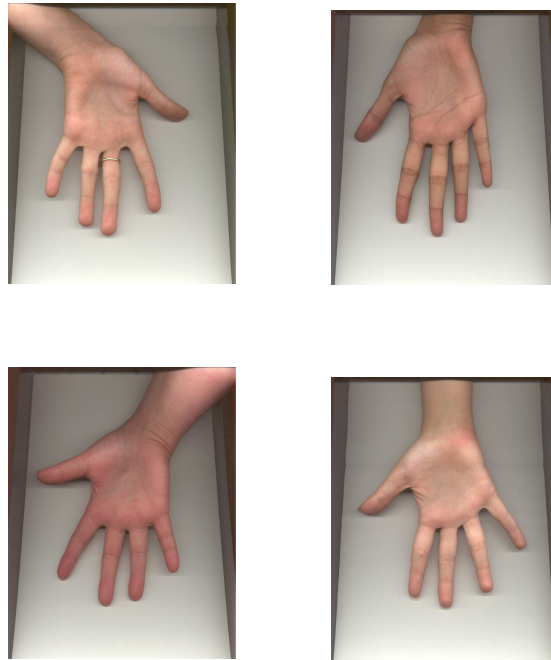


Fig. 10. Palmprint examples

- b) Graphical User Interface built with Visual Basic. NET technology to access "COLLECTION" database, Section IV-B.2.
- c) Biometric scanner interfaces that use device SDKs and are built with JAVA and Matlab, Section IV-B.3.
- d) Java application to generate XML files, Section IV-B.4.
- e) Bulk upload programs for every modality and collection (voice, face, fingerprint, iris, palmprint, hand and soft biometric such as weight and height) built with Java, Oracle XMLDB and PL/SQL, Section IV-B.5.

2) **VIA INTERNET** (work in progress)

A. *BIOMDATA DATABASE DESIGN*

The core of this project is "BIOMDATA" database, implemented with Oracle 10g database management systems [15] and pictured in Figure 12. As a first layer of data storage, the database is able to store images, video and sound files that are used for research in unimodal and multimodal biometrics. The second and third layers, the design for template information and match scores are in progress.

The novelty of our system is the multimodal property that allows us to gather biometric characteristics from different modalities for the same person from one or multiple data collection sessions using database management systems to store the biometric data. The initial design allows storing images and sound files for six biometric modalities: iris, voice, hand,

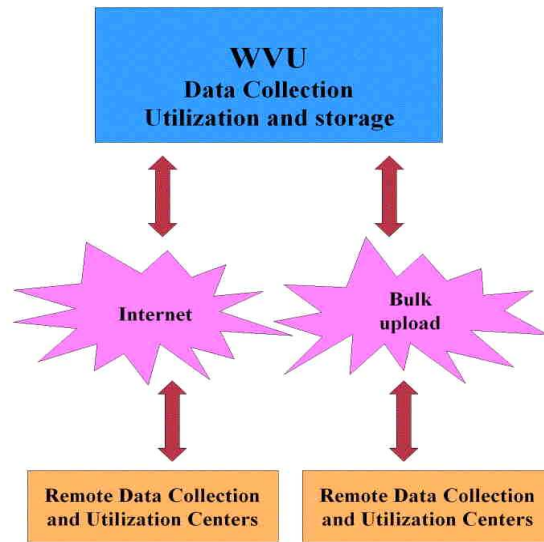


Fig. 11. Sharing WVU Multimodal Biometric Data

palmprint, fingerprint, face recognition and information related to soft biometrics: weight and height of the subjects. It is built in a flexible way that allows us to add new biometric modalities any time without interfering with the previous work. The multimodal biometric database is divided into five main groups. Common for every biometric modality are:

- Groups of data
- Participants
- Devices

Specific for every biometric modality:

- Variable factors
- Biometric Data

The bridge table **Record** is partitioned by biometric modality. Different *tablespaces* are defined for every modality. Images and sound files are stored on separate *tablespaces* per modality.

1) Groups of data

- Organizations
- Collections
- Sessions

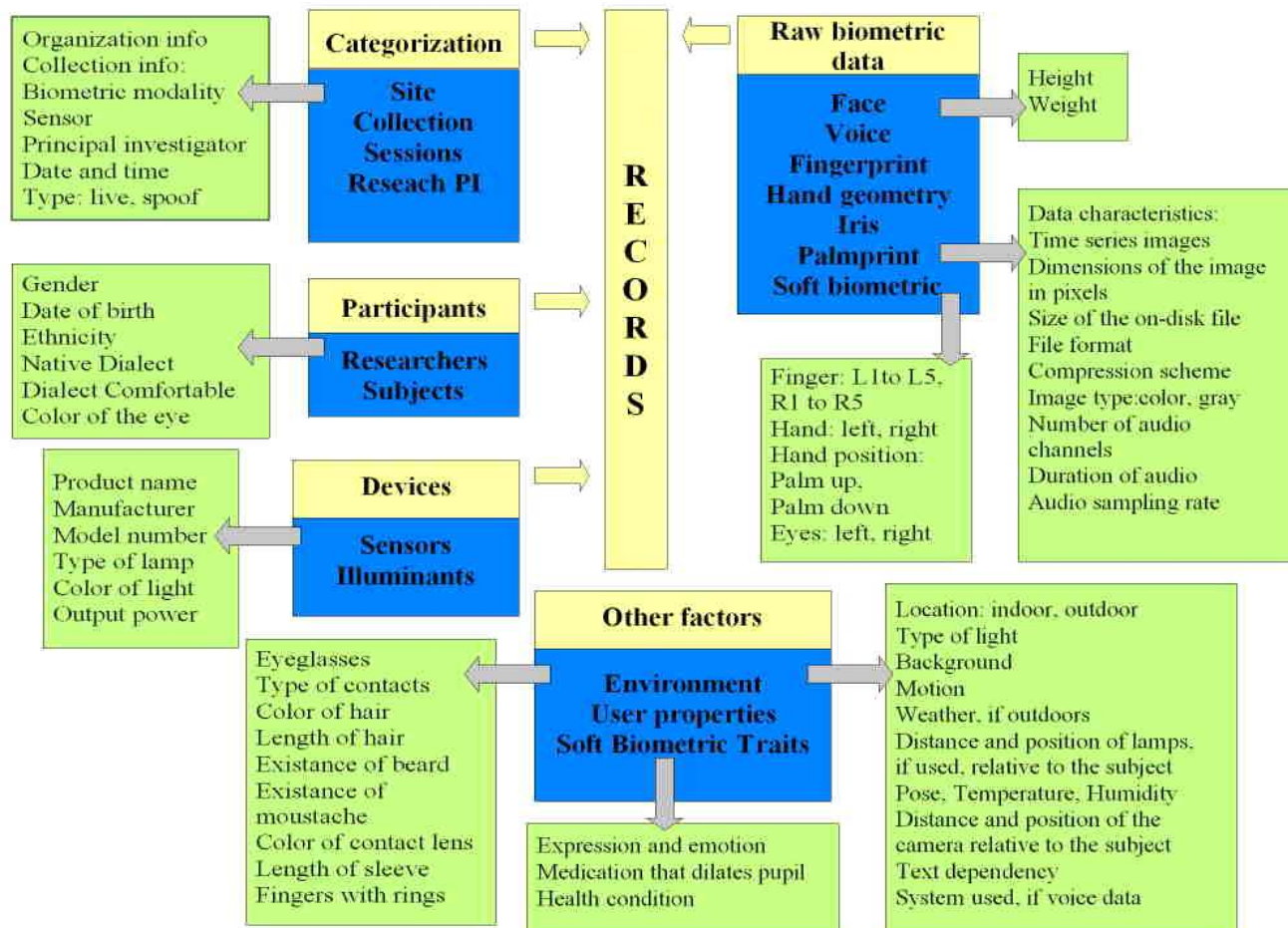


Fig. 12. The structure of the database

All data is grouped as per organization (site of data collection or department), collection (biometric modality and sensor within the project) and session. Database stores the name, department and the address of the organization; stores the name of the collection, the biometric modality, the sensor, the principal investigator, and the type (live, spoof or cadaver) of all the collections within organization. Sessions are determined by the date registered in the database at the time of data collection.

2) Participants

- Researchers
- Subjects

The database stores information regarding the researchers, name, email, telephone, title and the rank (if principal investigator or investigator for I.R.B. documents), that are in charge with the projects and can be contacted any time. For security and privacy issues, the name of the subjects (persons that decide voluntarily to participate in the project) is not

stored in the database. Every subject is allocated a random identification number, a seven digit number randomly generated in the database, in the first session of data collection and maintained in consecutive sessions for all biometric modalities. Every participant organization handles the random identification numbers. This number along with the organization's identification number is used to correlate sound or image files of different biometric features collected in the system for the same person. For statistical purpose the gender, date of birth, ethnicity, native and comfortable dialect, native color of the eye are stored for every subject. Subjects can be part of multiple collections and projects within the same organization. In case there's a method of payment, database tracks information regarding the number of sessions attended, date, remuneration and bonuses received by the subject.

3) **Devices**

- Sensors
- Illuminants

Database stores information about the devices: sensors and illuminants used, such as the name, model number, manufacturer, type, light color, and the computers on which they are installed within organization. Database design is flexible so that new sensors for any biometric modality and new illuminants can be added any time. According to the data collection protocol adopted, multiple sensors for the same biometric modality can be used in the same session; multiple images or sound recordings can be stored for the same sensor in the same session. As an example, for fingerprint recognition, for one subject, multiple fingerprint scanners can be used in one session, multiple fingerprint images can be captured for one and the same finger and fingerprints from multiple fingers can be captured in the same session. The design also allows storing images that contains fingerprints from multiple fingers in one image (for example all five fingers from one hand in one image) along with the fingers captured in the image.

4) **Variable factors**

- Environment
- Appearance
- Behavior

This group refers to the configurations used to collect data, to the appearance and expression of the subjects. These are specific for every biometric modality and described later in this section.

5) **Biometric Data**

- Face images
- Fingerprint images
- Hand geometry images
- Palmprint images

- Iris images
- Voice sound
- Soft biometrics (weight and height)

Biometric data refers to the biometric data itself: images and sound files for the six biometric modalities. We use Oracle InterMedia which is part of Oracle 10g, specifically Ordsys.OrdImage and Ordsys.OrdAudio [15], packages to store these files. This allows us to store additional information regarding biometric data that is useful in processing data.

For image files:

- The height and width of the image in pixels
- The size of the *on-disk* image file in bytes
- File type or format in which the image data is stored
- Compression format
- Type of image (if color or grayscale)

For sound files:

- Format in which the audio data is stored
- Number of channels
- Sampling rate
- Compression type
- Audio duration

1) Description of data for individual modalities:

a) *Face recognition*: Every face image stored into the database is associated with the configuration used to capture the picture of the face, the physical appearance of the subject and the expression and emotion of the subject. The design incorporates multiple situations (a)single image captures, when the face and the camera are still or (b)time series image captures,(b1) when the face is still and the camera is moving or (b2)the face is moving, tracking an object and the camera is still. The following environment data is stored:

- Environment - capture location of the image: indoor, outdoor
- Time series capture or single capture
- Light - refers to the type of external lighting: natural, lamps, infrared etc.
- Background - type of background used: cluttered or uncluttered
- Motion - refers to the movement of the head: still or tracking an object
- Weather - if the pictures are taken outdoors, weather conditions are recorded in the database
- Lamps - if the pictures are taken indoors, illuminants used, distance and position of the lamps relative to the subject are recorded
- Pose - refers to the yaw, pitch and roll, the angles in the three planes, for the position of the face; the facial pose of subjects in still imagery. For time series pictures the entire sequence of values for pose is stored in advanced in the

database.

- Position of the camera relative to the frontal face. If camera is moving the sequence of angles is stored in the database.
- Distance - distance of the camera to the subject
- Unit of measure for the distance

Configuration for face recognition can be applied to video capture of the face image. During data collection, the subject can change the appearance from one session to another. The subject can be asked to change also the expression of the face. The following information is saved into the database for every data collection session:

- Wearing eyeglasses or the type of contacts
- Color of the hair
- Length of the hair
- Existence of the beard
- Existence of the moustache
- Color of the eye, that can be changed using contacts
- Expression and emotion

The database registers also the use of colorchecker board in the picture.

b) Fingerprint recognition: Fingerprint recognition modality necessitates different environment data to be stored:

- Environment temperature
- Unit of measure for temperature
- Environment humidity
- Unit of measure for humidity
- Finger, from L1 to L5, and R1 to R5, or enumeration of fingers.

c) Hand geometry: Research in hand geometry recognition requires the following additional information:

- Left or right hand
- Palm up or palm down
- Rings - enumeration of the fingers on which the subject has rings. Fingers are coded with the letter L or R, L for the left hand and R for the right hand, and the number of the finger, starting with 1, the thumb up to 5, the pinky.
- Sleeve, long or short - refers to the length of the sleeve. The sleeve can cover part of the hand.

d) Iris recognition: Projects for iris recognition modality are divided into two main groups: non-ideal iris and off-angle iris recognition. Non-ideal iris recognition implies capture of the iris from different positions and distances; eye can be partially occluded. Off-angle iris recognition implies capture of the iris from specific angles and distances. The design incorporates multiple situations for single image captures, when the eye and the camera are still or time series image captures, when the eyes are still, gazing to the same object and the camera is moving or the eyes are moving, tracking an object and the camera is still. The following environment data is stored:

- Environment - capture location of the image: indoor, outdoor

- Time series capture or single capture
- Weather - if the pictures are taken outdoors, weather conditions are stored into the database
- Lamps - if the pictures are taken indoor, illuminants used, distance and position of the lamps relative to the subject are recorded.
- Pose - refers to the yaw, pitch and roll, the angles in the three planes, for the direction of the eyes relative to the face.

For time series pictures the entire sequence of values for pose is stored in the database.

- Position of the camera or iris scanner relative to the straight look. If camera is moving the sequence of angles is stored in the database
- Distance of the camera or iris scanner to the subject
- Unit of measure for the distance
- Motion - refers to the movement of the eyes: still or tracking an object
- Left, right or both eyes
- Use of medication that can influence the dilation of the pupil.

e) *Voice recognition*: Among environmental factors stored for voice recognition there are:

- The text dependency of recording
- Number of speakers
- Language used
- Location where voice is recorded: outdoors or indoors
- System used such as telephone, microphone, and wireless

B. DATA COLLECTION PACKAGE FOR BULK UPLOAD

1) *COLLECTION database design*: *COLLECTION* database (Microsoft Access) resides in one of the computers (Pentium IV, Microsoft XP) from the Multimodal Biometric Dataset Collection Laboratory, Figure4 a, b. It is used to store soft biometric data (weight, height), and additional information regarding the collections, configurations, appearance of the subjects and environment conditions when data is collected. Even though the design of the two databases is different, table fields from *COLLECTION* database have a correspondent table field in *BIOMDATA* database. For bulk upload, part of the information that needs to be inserted into *BIOMDATA* and is not in *COLLECTION* database, is found in the naming file convention described in Section IV-B.6. Within modality data is divided by collections, sessions and records. Collections are defined per sensor, type of data (live, cadaver, spoof) and configuration or environment factors. In order to keep data in the database consistent, same check constraints on table columns are imposed in both databases; restricted values on the columns of the tables are provided by the GUI used to populate the Collection database, through combo boxes. The database contains also multiple queries called by Java application when an XML file is created.

2) *Graphical User Interface for COLLECTION database*: This application is built with Visual Basic.Net technology and is used to insert data into the *COLLECTION* database, Figure 13. The main data collection screen consists of multiple frames: Subject Random ID and Subject Data frames through which a random identification number is assigned for a new subject; if the subject is a repeat, his/her random identification number is verified and confirmed with date of birth and gender displayed in a message window. For the new subjects, in the first session, date of birth, ethnicity, gender, native dialect, comfortable dialect, and color of the eye are registered for statistical purpose. Frames like Face Data, Finger Data, Iris Data, Hand Data, Voice Data, Palmprint Data, Soft Biometrics gather additional information for each modality: type of contact lenses (if any), color of the hair, length of the hair, existence of the beard or moustache, color of the eyes, temperature and humidity of the environment where data is acquired, along with the unit of measure for temperature and humidity, status of the finger (clean, dirty, wet, dry etc.), medication in the eyes (if any), length of the sleeves, the fingers with rings, health of the subject when data is acquired (healthy, running nose, sore throat, influenza, etc.), weight and height of the subject along with the unit of measure for the weight and height. By pressing the SaveData command button from the Command Menu frame the data is inserted into the database.

3) *Biometric Scanner Interface*: For hand geometry we use a scanner interface developed in Matlab, and supplied by IR Recognition Systems. We collect the images of the hand along with the feature vectors in a log file.

For fingerprint biometric scanner we use our own program implemented with JAVA that uses the SDK provided by the constructor of the fingerprint biometric scanner. This interface allows us to capture single or time series images at different adjustable intervals of time.

4) *Generate XML files*: In order to upload the biometric images and sound files along with the information from the *COLLECTION* database into the Oracle database, we correlate the information from the *COLLECTION* Access database with the biometric data that is collected for every modality by using specific naming conventions for the biometric files described in section IV-B.6. Based on the information from the *COLLECTION* database, the Java application generates XML files. Each XML file is per modality and per collection within modality.

The Major Java API used for creation of xml files is **JDOM: Java Document Object Model**. JDOM is a new open source API for reading, writing, and manipulating XML from within Java code. JDOM interoperates well with existing standards such as the Simple API for XML (SAX) and the Document Object Model (DOM). JDOM attempts to incorporate the best of DOM and SAX. It's a lightweight API designed to perform quickly in a small-memory footprint. JDOM also provides a full document view with random access but, surprisingly, it does not require the entire document to be in memory. The API allows for future flyweight implementations that load information only when needed. Additionally, JDOM supports easy document

Collection

☐ Face Enable/Disable
 ☐ Finger Enable/Disable
 ☐ Iris Enable/disable

Current Date
12/7/2006 4:32:40 PM

Subject Random ID
RandomID
[Text Field]
Check subject

Subject Data

Gender [Dropdown]

Date of Birth
Month [Dropdown]
Day [Dropdown]
Year [Dropdown]

Ethnicity [Dropdown]

Dialect Native [Text Field]

Dialect Comfortable [Text Field]

Eye Color [Dropdown]

Miscellaneous [Text Area]

Clear Subject Data

Face Data

Lens [Dropdown]

Hair Color [Dropdown]

Hair Length [Dropdown]

Beard [Dropdown]

Moustache [Dropdown]

Eye Color [Dropdown]

Collection [Text Area]

Miscellaneous [Text Area]

Clear Face Data

Finger Data

Temperature [Text Field]

UM Temperature [Dropdown]

Humidity [Text Field]

UM Humidity [Dropdown]

Finger Status
Clean
Perspired
Wet
Dry
Dirty

Collection [Text Area]

Miscellaneous [Text Area]

Clear Finger Data

Iris Data

Eye Color [Dropdown]

Medication [Dropdown]

Lens [Dropdown]

Collection [Text Area]

Miscellaneous [Text Area]

Clear Iris Data

☐ Hand Enable/Disable
 ☐ Voice Enable/Disable
 ☐ Soft Biometric Enable/Disable
 ☐ Palm Enable/Disable

Hand Data

Sleeve [Dropdown]

Rings [List Box: L1, L2, L3, L4, L5, R1, R2]

Collection [Text Area]

Miscellaneous [Text Area]

Clear Hand Data

Voice Data

Health [Dropdown]

Collection [Text Area]

Miscellaneous [Text Area]

Clear Voice Data

Soft Biometrics

Weight [Text Field]

UM weight [Dropdown]

Height [Text Field]

UM height [Dropdown]

Collection [Text Area]

Miscellaneous [Text Area]

Clear Soft Data

Palm Data

Sleeve [Dropdown]

Rings [List Box: L1, L2, L3, L4, L5]

Collection [Text Area]

Miscellaneous [Text Area]

Clear Palm Data

Command Menu

Save Data **Exit Data Collection**

Fig. 13. Graphical User Interface for COLLECTION Database

modification through standard constructors and normal set methods.

5) *Bulk upload programs*: The bulk upload program parses the XML file, gathers the information that is to be inserted, searches for the appropriate image or sound files from the bulk and inserts all the information into *BIOMDATA* database. For bulk upload into the Oracle database we use Oracle XMLDB. XML is a text-based markup language that is fast becoming the standard for data interchange on the Web. As with HTML, you identify data using tags (identifiers enclosed in angle brackets). Collectively, the tags are known as "markup". But unlike HTML, XML tags identify the data, rather than specifying how to display it. An XML tag acts like a field name in the program, and is identical with the field name from the database tables. It puts a label on a piece of data that identifies it (for example: `< message > data < /message >`). The Java API used for parsing the XML data is **SAX: Serial Access with the Simple API for XML**.

SAX is an open source API which uses serial-access mechanism for accessing XML documents. This is the protocol that most servlets and network-oriented programs will want to use to transmit and receive XML documents, because it's the fastest and least memory-intensive mechanism that is currently available for dealing with XML documents. It is an event-driven model.

6) *Naming conventions for the biometric image and sound files*: For all naming conventions, the following notations apply:

- YYYY - The four-digit year of the data collection
- MM - The number of the month in the year, from 1 through 12 of the data collection
- DD - The day in the month, from 1 through 31, of the data collection
- HH24 - The hour in the day, from 0 through 23
- MI - The minute component of the date's time, from 0 through 59
- SS - The second component of the date's time, from 0 through 59
- xxx - The millisecond component of the date's time, from 0 through 999 of the data collection. Mentioned only when necessary.
- RandomID - The random identification number, a seven digit number assigned in the first session of data collection for every subject
- Modality - The biometric modality, a two letter convention: FG - fingerprint, IR - iris, VC - voice, FC - face, PL - palmprint, HD - hand
- SequenceNumber - The number used to name the sequence in the time series images. ImageNumber - The number of the single image collected in the same session.
- DeviceNr - The identification number of the biometric device or scanner.
- Org - The organization number.

a) Naming file convention for face images:

RandomID_YYYYMMDDHHMMSSxxx
_SequenceNumber_Modality
*+ DeviceNr_Org.**

Example: 1112164_20040521002800_3_FC12_00.bmp

- b) Naming file convention for fingerprint images:

RandomID_YYYYMMDDHHMMSSxxx

_Finger_SequenceNumber

*_ModalityDeviceNr_Org.**

where Finger is: L1, L2, L3, L4, L5 (left hand, starting from thumb to pinkie) or R1, R2, R3, R4, R5 (right hand, starting from thumb to pinkie) or any other combination as L1L2, L1L2L3L4L5 etc.

Example: 1112164_20042305201420_R1_0_FG07_00.bmp

- c) Naming file convention for iris images:

RandomID_YYYYMMDDHH24MISS

_Eye_SequenceNumber

*_Modality + DeviceNr_Org.**

where Eye is: R- right eye, L- left eye or B- for both eyes A supplementary notepad file with extension .log that contains written data collected from the image capture is also saved with the same name, different extension. Biometric scanners have different naming file rules. This problem is being addressed by creating a DOS batch code that renames the image to fit the convention of the rest of the data.

Example: 1112164_20040521120704_L_0_IR06_00.bmp

- d) Naming file convention for hand images:

RandomID_YYYYMMDDHH24MISS

_Hand_ImageNumber

*_Modality + DeviceNr_Org.**

where Hand is: L- for left hand and R- for right hand. A .txt file is created at the time of data collection with feature vectors and match score with the same name.

Example: 1112164_20040414180827_R_HD05_00.jpg

- e) Naming file convention for palmprint images:

RandomID_YYYYMMDDHH24MISS

_Palm_ImageNumber

*_Modality + DeviceNr_Org.**

where Palm is: L for left hand and R for right hand.

Example: 1112164_20040727110900_R_0_PL04_00.bmp

- f) Naming file convention for voice sound:

RandomID_YYYYMMDDHH24MISS

_SoundNumber_Modality

*+ DeviceNr_Org.**

Example: 1112164_20040414181900_1_VC14_00.wav

V. DISTRIBUTION

With some restrictions, WVU Multimodal Biometric Database is available and distributed freely to the biometric research community. Institutions willing to obtain the database will have to sign the Database Release Agreement and pay for the shipping of the CDs or DVDs.

Contact information: Lawrence.Hornak@mail.wvu.edu or Arun.Ross@mail.wvu.edu

VI. CONCLUSIONS AND PERSPECTIVES

In this paper, a large multimodal biometric database with its associated protocols and an entire system to store such information using relational database management systems, have been presented. WVU multimodal biometric database offers the opportunity for research community to test and evaluate the performance of their algorithms and moreover, the chance to exploit the benefits of multimodal biometrics by data fusion. Up to now, the first release is already distributed to different institutions. Besides the modalities mentioned when biometric data was presented in this paper, the second release, which is in the process of inspection for quality, contains video face with voice, and fingerprint images collected from different fingerprint sensors. In the future, new sessions with the same and new people will be acquired.

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